

Cyber Risk & Threat Intelligence System



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1. Abstract

Cybersecurity threats are increasing rapidly due to the widespread use of networked systems and internet-based services. Traditional security monitoring approaches often struggle to identify sophisticated attacks in real time. This project presents a Cyber Risk & Threat Intelligence System that combines Machine Learning (ML), rule-based reasoning using Prolog, packet analysis, encrypted threat storage, and automated incident reporting. The system analyzes network traffic from PCAP files, extracts packet features, detects suspicious activities, classifies threats using a trained machine learning model, and stores incident records securely using AES-256-GCM encryption. A web-based dashboard provides real-time monitoring and visualization of detected threats. The proposed solution improves threat detection, incident management, and security intelligence while maintaining confidentiality and integrity of sensitive security logs.

2. Introduction

The increasing dependence on digital infrastructure has led to a rise in cyberattacks such as intrusion attempts, malware infections, network scans, SQL injection attacks, and denial-of-service attacks. Organizations require intelligent systems capable of monitoring network traffic and identifying malicious activities automatically.

This project develops a Cyber Risk & Threat Intelligence System that integrates Artificial Intelligence techniques with cybersecurity principles. The system captures and analyzes network traffic, applies machine learning models for anomaly detection, uses Prolog-based reasoning for threat diagnosis, securely stores incident information using encryption, and generates incident reports for security analysts.

The project demonstrates how AI can be applied in practical cybersecurity environments to improve threat detection and response capabilities.

3. Problem Statement

Modern networks generate a large volume of traffic, making manual monitoring difficult and inefficient. Security analysts often face challenges in:

- Detecting malicious traffic in real time.
- Identifying suspicious network behavior.
- Managing large amounts of security logs securely.
- Generating incident reports efficiently.
- Combining automated detection with expert reasoning.

Therefore, there is a need for an intelligent threat intelligence system that can automatically analyze network traffic, detect cyber threats, classify risks, securely store evidence, and provide meaningful security insights.

4. Background

Several cybersecurity solutions use machine learning for intrusion detection and anomaly detection. Datasets such as NSL-KDD are commonly used for training classification models that distinguish between normal and malicious network activities.

Research shows that:

- Machine Learning improves intrusion detection accuracy.
- K-Nearest Neighbors (KNN) is effective for network traffic classification.

- Rule-based systems provide explainable threat diagnosis.
- AES-256 encryption is widely adopted for secure storage of sensitive information.
- Threat intelligence platforms help organizations respond to cyber incidents more efficiently.

This project combines these concepts into a single integrated platform.

5. Methodology

The system follows the workflow shown below:

1. Upload or capture network traffic.
2. Extract packet features.
3. Perform machine learning classification.
4. Apply rule-based threat reasoning.
5. Calculate risk level.
6. Encrypt and store threat logs.
7. Generate incident reports.
8. Display results on dashboard.

5.1.1 Development Tools

- Python
- Flask
- Scikit-learn
- PyShark
- SQLite
- SWI-Prolog
- Flask-SocketIO
- AES-256-GCM Encryption

5.1.2 Dataset

The system uses NSL-KDD based datasets:

- nsl_kdd_large.csv
- nsl_kdd_synthetic.csv

These datasets are used to train and evaluate the machine learning model.

6. Algorithm Explanation

6.1. Step 1: Data Collection

Network packets are obtained from:

- Uploaded PCAP files
- Live packet capture

6.2. Step 2: Feature Extraction

Important features extracted include:

- Source IP Address
- Destination IP Address

- Protocol Type
- Source Port
- Destination Port
- Packet Length
- Payload Information

6.3. Step 3: Data Preprocessing

- Missing values removed
- Categorical features encoded
- Features normalized using StandardScaler

6.4. Step 4: Model Training

The project primarily uses the K-Nearest Neighbors (KNN) algorithm.

6.4.1 KNN Algorithm

1. Store training samples.
2. Calculate distance between test sample and training samples.
3. Select K nearest neighbors.
4. Assign the most frequent class among neighbors.

K = 5 was used in the implementation.

6.5. Step 5: Threat Classification

The trained model predicts whether traffic is:

- Normal
- Malicious
- Anomalous

6.6. Step 6: Rule-Based Reasoning

Prolog knowledge base applies expert security rules to determine:

- Threat category
- Severity
- Recommended mitigation actions

6.7. Step 7: Secure Storage

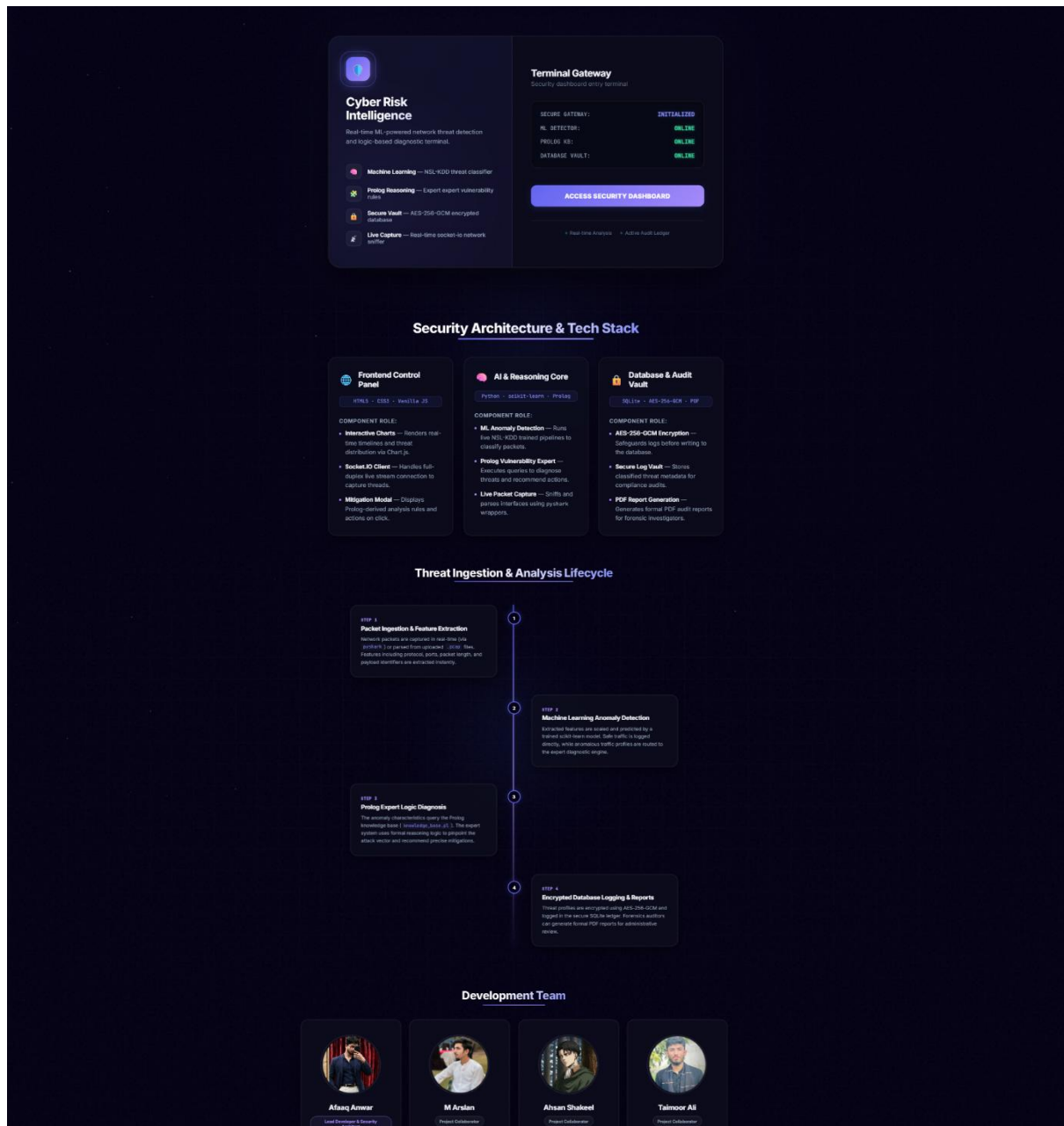
Threat logs are encrypted using AES-256-GCM before storage in SQLite database.

6.8. Step 8: Report Generation

The system automatically generates incident reports for detected threats.

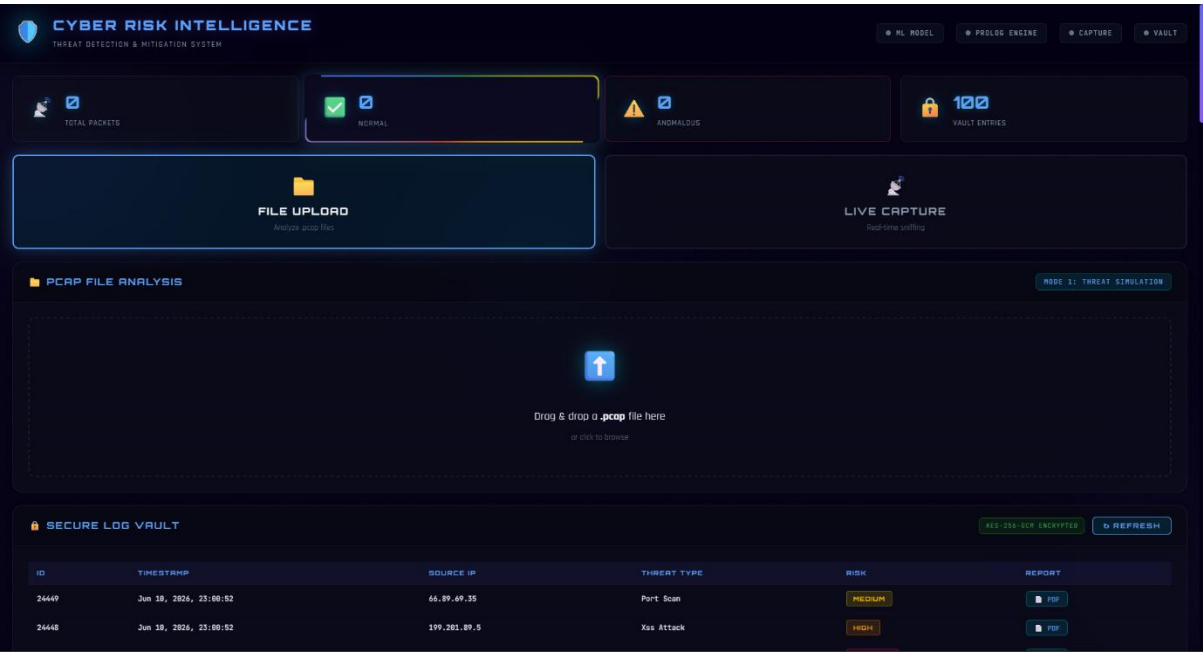
7. Results and Screenshots

7.1.1 Screenshot 1: Landing Page

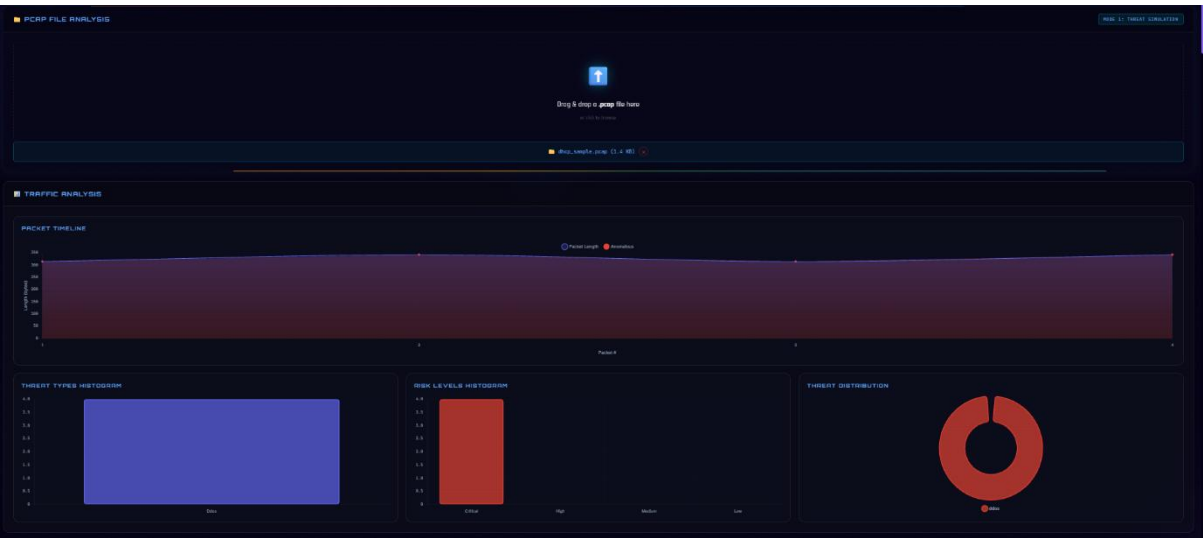


7.1.2 Screenshot 2: Dashboard Interface

Cyber Risk & Threat Intelligence System



7.1.3 Screenshot 3: PCAP File Upload



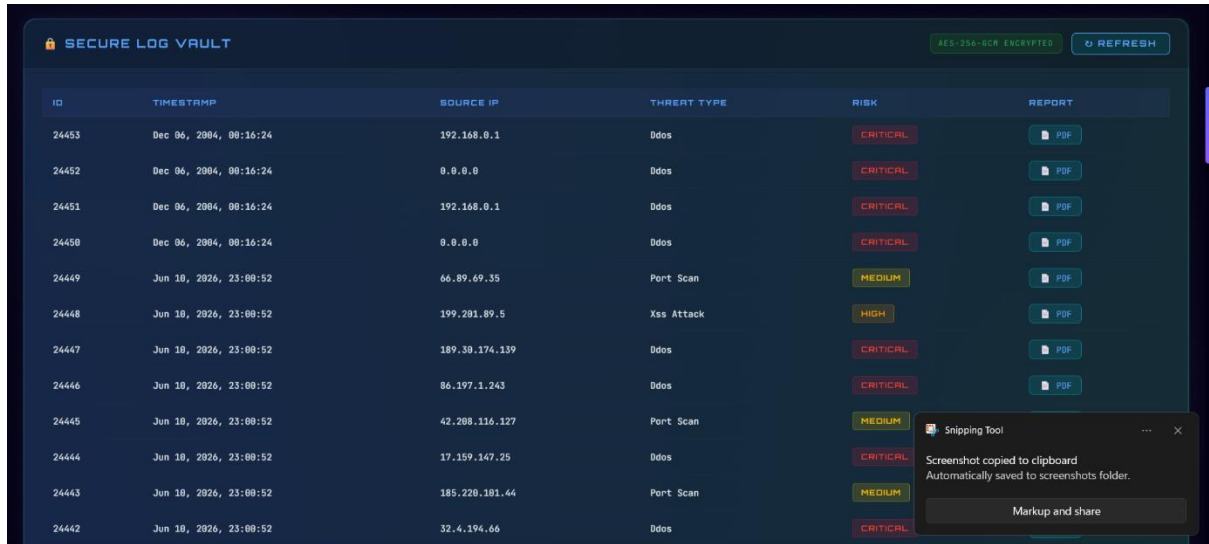
7.1.4 Screenshot 4: Threat Detection Result

The Threat Analysis Results section displays a table of detected threats. The table includes columns for #, SOURCE IP, DEST IP, PROTOCOL, PORT, THREAT TYPE, RISK, CONFIDENCE, and ACTION. The table shows four entries, all identified as Ddos attacks with a CRITICAL risk level and 60.0% confidence. The ACTION column provides a 'View' link for each entry.

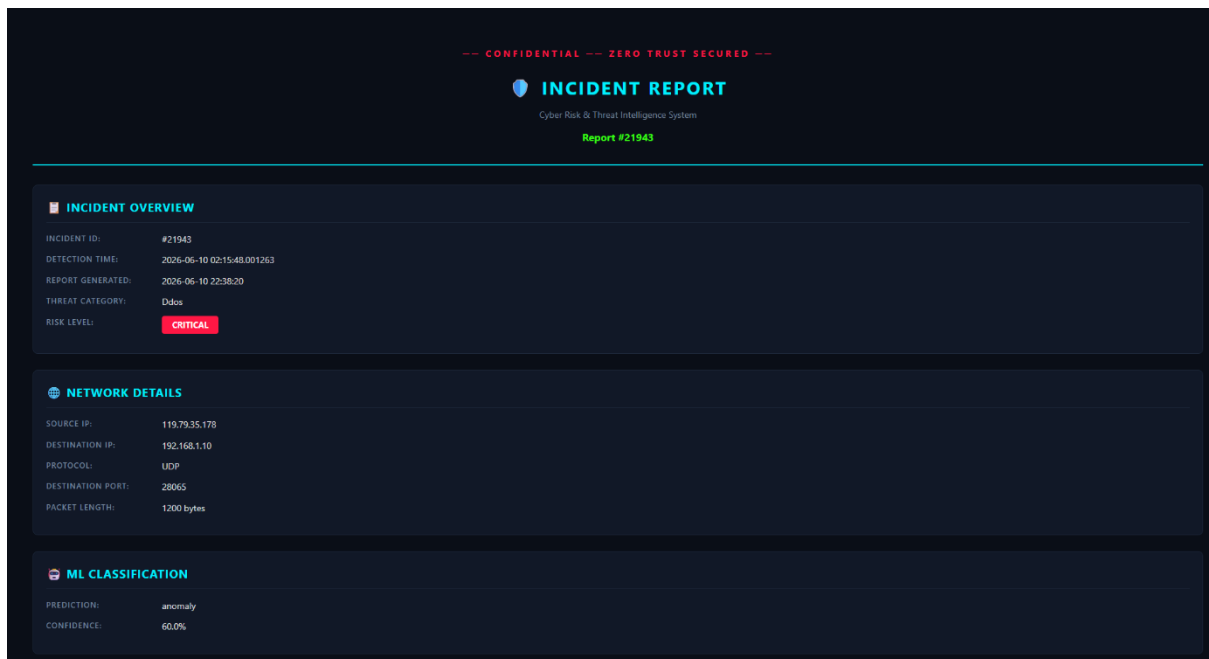
#	SOURCE IP	DEST IP	PROTOCOL	PORT	THREAT TYPE	RISK	CONFIDENCE	ACTION
1	0.0.0.0	255.255.255.255	UDP	67	Ddos	CRITICAL	60.0%	View
2	192.168.0.1	192.168.0.19	UDP	68	Ddos	CRITICAL	60.0%	View
3	0.0.0.0	255.255.255.255	UDP	67	Ddos	CRITICAL	60.0%	View
4	192.168.0.1	192.168.0.19	UDP	68	Ddos	CRITICAL	60.0%	View

7.1.5 Screenshot 5: Threat Logs

Cyber Risk & Threat Intelligence System



7.1.6 Screenshot 6: Generated Incident Report



7.1.7 Sample Results

Parameter	Result
Dataset Used	NSL-KDD
Model	KNN
K Value	5
Threat Detection	Successful
Report Generation	Successful
Secure Log Storage	Successful
Real-Time Monitoring	Supported

8. Evaluation

Cyber Risk & Threat Intelligence System

The system was evaluated based on functionality and performance.

8.1.1 Evaluation Metrics

- Threat Detection Accuracy
- Classification Reliability
- Encryption Security
- System Response Time
- Report Generation Capability

8.1.2 Observations

- Successfully classified network traffic.
- Detected suspicious packet patterns.
- Stored logs securely using AES encryption.
- Generated readable incident reports.
- Provided real-time dashboard monitoring.

The integrated approach improved both security monitoring and incident management.

9. Ethical Considerations

The project was developed with responsible cybersecurity practices.

- Used only educational and research datasets.
- No unauthorized network monitoring was performed.
- User privacy was protected.
- Threat logs were encrypted to maintain confidentiality.
- The system is intended for defensive cybersecurity purposes only.

10. Limitations

The current implementation has several limitations:

- Depends on the quality of training data.
- KNN performance may decrease with very large datasets.
- Limited threat categories compared to enterprise solutions.
- Prolog rules require manual updates for new attack types.
- Real-world deployment would require larger datasets and advanced models.

11. Conclusion

This project successfully developed an AI-powered Cyber Risk & Threat Intelligence System capable of detecting suspicious network activities, classifying threats using machine learning, applying rule-based reasoning, securely storing threat information, and generating incident reports. The integration of KNN classification, Prolog reasoning, AES-256 encryption, and Flask-based visualization demonstrates the practical application of Artificial Intelligence in

cybersecurity. The system provides an effective foundation for intelligent network monitoring and cyber threat analysis.

12. References

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